PLC Internal Operations & Programming

1 The Processor (CPU)

CPU: The Brain of the PLC

The **Central Processing Unit (CPU)** controls all PLC activities. It consists of two primary functional areas:

1. **Microprocessor:** The ”engine” that implements control logic and manages communication between modules.
2. **Memory:** A digital storage area for:
   - User program instructions.
   - Numerical values for timers and counters.
   - Real-time status of I/O devices.

1.1 The PLC Scan Cycle

The CPU operates in a continuous, high-speed loop called the **PLC Scan**. The sequence is as follows:

Step	Activity	Description
1	**Read Inputs**	CPU checks the status of all input field devices.
2	**Execute Program**	CPU runs the application logic (e.g., Ladder Logic).
3	**Update Outputs**	CPU sets the status of all output field devices based on the program result.
4	**Diagnostics**	CPU performs internal self-checks and communication tasks.

2 The I/O System

I/O Interface

The **I/O system** acts as the interface between the PLC’s internal low-voltage circuitry and the ”real-world” field devices.

- **Field Devices:** These are the external sensors (Inputs) and actuators (Outputs) hardwired to the PLC terminals.
- **Signal Conditioning:** The I/O modules condition incoming signals from sensors and outgoing signals to actuators.

2.1 Optical Isolation

Optical Isolators

To protect the sensitive microprocessor from high-voltage spikes or electrical noise on the factory floor, PLCs use **Optical Isolators**.

- They use light to "couple" circuits together without a direct electrical connection.
- This completely isolates the internal CPU components from the external I/O terminals.

3 Hardware and Programming Tools

3.1 Power Supply

The power supply provides the required DC power to modules plugged into the rack.

- **Large Systems:** Field devices usually require separate, external AC or DC power sources.
- **Small Micro PLCs:** The primary PLC power supply might be capable of powering field devices directly.

3.2 Programming Devices

These are used to enter the control logic into the PLC's memory.

- **Hand-held Devices:** Inexpensive and portable; ideal for programming small PLCs or making quick field edits.
- **Personal Computer (PC):** The most common tool. Using specialized software, users can create, edit, document, and troubleshoot programs.
  - **Connectivity:** Communicates via Serial, Parallel, or **Ethernet**.

4 Programming Languages

Relay Ladder Logic (RLL)

Relay Ladder Logic is the standard programming language for PLCs. It is a graphical language that represents logic as a series of "rungs" containing contacts and coils.

Key Origins & Features:

- **Origin:** Derived from electromechanical relay control schematics.

- **Graphic Symbols:** Uses symbols (like normally open/closed contacts) that represent their intended physical outcome.
- **Structure:** Visually represents logic like a ladder, where each rung is a specific instruction or action.

The speed of the scan cycle determines the responsiveness of the entire industrial process.